Computing endomorphisms of superspecial principally polarized abelian surfaces in time and memory $p^{4/5+o(1)}$

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Abstract. We present an algorithm for computing endomorphisms of superspecial principally polarized abelian surfaces over finite fields. Under a standard heuristic, we obtain a non-scalar polarized endomorphism in expected time and memory $p^{4/5+o(1)}$.

1 Introduction

Let p be a prime number and let (A, λ_A) and (C, λ_C) denote maximal superspecial principally polarized abelian surfaces over $\mathbb{F}_{p^2}$. Recall the following computational problems (our descriptions are deliberately informal for the moment).

- OneEnd. Compute a polarized endomorphism $\alpha \in \mathrm{End}(A) \setminus \mathbb{Z}$.
- EndRing. Compute a $\mathbb{Z}$-basis of the endomorphism ring $\mathrm{End}(A)$.
- Isogeny. Compute a polarized isogeny $\varphi : A \to C$.

These problems constitute the two-dimensional variants of the fundamental computational problems underlying isogeny-based cryptography for elliptic curves. The best known classical algorithms all have time complexity $\tilde{O}(p)$ (for any one of these problems). The first heuristic algorithm for Isogeny in time $\tilde{O}(p)$ was given by Costello–Smith [CS20]: it reduces the problem to the (easier) one-dimensional case by performing a random walk in the $(2,2)$-isogeny graph until landing on a product of elliptic curves $E_1 \times E_2$. The bottleneck in this approach is the first step, which takes expected time $\tilde{O}(p)$. This approach is conditional on the heuristic that $(2,2)$-isogeny walks exhibit *rapid mixing*; the latter was later proved by [ATY24] for lazy walks. Under the mixing hypothesis, in [ABI+25, Cor. 5.20] it is moreover shown how to recover EndRing in $\tilde{O}(p)$. A recent paper [SKTW26] proves significantly sharper bounds on spectral gaps of isogeny graphs, and gives an unconditional algorithm for EndRing in dimension $g > 1$ in expected time $\tilde{O}(p^{g-1})$.

A significant practical improvement to Costello–Smith is SplitSearcher [SCF24]. In an extended version [SCF26] of the same paper, the authors also describe a potential alternative approach to Costello–Smith: instead of walking to a product surface, one could walk to a surface with real multiplication, for some RM order of fixed discriminant D. Although they do not claim a complexity estimate for an attack based on this approach, the density of such surfaces is

heuristically $\Theta(p^{-1})$ (with the implicit constant now depending on D), which, with efficient RM detection, suggests an expected search time of $\tilde{O}(p)$. We also mention another recent work [Sha26], which gives explicit asymptotic formulas for the density of superspecial surfaces admitting optimal (N, N)-splittings at fixed levels N and gives counts for optimal RM structures consistent with the predicted $\Theta(p^{-1})$ density of RM surfaces [Sha26, Cor. 9.2].

Main results

Our first result is an improvement for OneEnd. Let $p > 3$ be a prime number such that $p \equiv 3 \pmod 4$, and let (A, λ_A) be a maximal superspecial principally polarized abelian surface over $\mathbb{F}_{p^2}$.

Theorem 1.1. *Assuming Heuristic 1, there is a randomized algorithm that computes an efficient representation of a polarized endomorphism $\alpha \in \mathrm{End}(A) \setminus \mathbb{Z}$ in expected time and memory $p^{4/5+o(1)}$.*

The underlying heuristic is a standard assumption on the smoothness probability of the multiplier of a random polarized isogeny. It is essentially the same heuristic driving Wesolowski's $O(p^{1/3+o(1)})$ attack for the one-dimensional isogeny problem [Wes26]. We provide experimental evidence supporting this heuristic in Section 9. A proof-of-concept Magma implementation can be found in the git repository associated to this paper.[1]

Under an additional heuristic assumption regarding the solutions to the OneEnd problem given by the algorithm, we also establish the following strong version of EndRing.

Theorem 1.2. *Assuming Heuristics 1 and 2, there is a randomized algorithm that computes a $\mathbb{Z}$-basis for $\mathrm{End}(A)$ consisting of efficiently represented endomorphisms, a multiplication table for the associated ring structure, and a matrix representation for the Rosati involution, in expected time and memory $p^{4/5+o(1)}$.*

Using the machinery of [PRS26], we can further extend this result to a solution to the Isogeny problem, conditional on the Generalized Riemann Hypothesis (GRH). Let (C, λ_C) denote another maximal superspecial principally polarized abelian surface over $\mathbb{F}_{p^2}$.

Theorem 1.3. *Assume Heuristics 1 and 2 and GRH. Then we can compute an efficient representation of a polarized isogeny $A \to C$ in expected time and memory $p^{4/5+o(1)}$.*

Technical overview

The main mechanism behind Theorem 1.1 comes from a recent result by Pham [Pha26, Prop. 4.1], which implies that the shortest nonzero vector α of the rank-five sublattice of trace-zero self-adjoint endomorphisms

$$\mathscr{S}_A := \{x \in \mathrm{End}(A) \mid x^\dagger = x,\ \mathrm{Trd}(x) = 0\}$$

[1] `https://github.com/houbenmr/attack-four-fifth`

has multiplier at most $X := \lfloor (p^2/2)^{1/5} \rfloor$. We further show (Theorem 3.3) that α always corresponds to an (n, n)-endomorphism with multiplier $n \leq X$. We then proceed, after randomizing A via a $(2, 2)$-isogeny walk, with a meet-in-the-middle attack: enumerate all outgoing (m, m)-isogenies from A with multiplier $m \leq X^{1/2+o(1)}$. Similar to Wesolowski's attack [Wes26], we impose a subexponential smoothness bound on m. We then assume (Heuristic 1) that n satisfies this smoothness bound with the probability one would expect from a random integer of the same size. We then prove a bound on the total number N of (m, m)-subgroups of A, summed over all $m \leq Y$; in particular, we show that $N \leq Y^4$ (Proposition 4.4). Setting $Y = X^{1/2+o(1)}$, this then bounds the expected time and memory complexity of the meet-in-the-middle attack by $X^{2+o(1)} = p^{4/5+o(1)}$.

To obtain algorithms for EndRing and Isogeny, we prove that $\mathscr{S}_A$ generates the full endomorphism ring (as a ring); see Theorem 7.1. Under an additional assumption (Heuristic 2) about the distribution of the outputs of a variant of the algorithm of Theorem 1.1 that is restricted to return only elements of $\mathscr{S}_A$, we show we can obtain a basis for $\mathscr{S}_A$ through a lattice saturation argument; this argument, contrary to what one would get from an arbitrary OneEnd oracle, relies on the special property of $\mathscr{S}_A$ that nonzero integral linear combinations and their integral halves remain in $\mathscr{S}_A$ and are therefore automatically polarized, allowing polarized isogeny interpolation throughout the saturation procedure.

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2 Notations and preliminaries

Unless explicitly mentioned otherwise, (A, λ_A) and (C, λ_C) will denote maximal superspecial principally polarized abelian surfaces over a finite field $\mathbb{F}_{p^2}$, where $p > 3$ is a prime number that is $\equiv 3 \pmod 4$. We will often drop the polarization from the notation, and also say $A/\mathbb{F}_{p^2}$ is a maximal superspecial *ppas*. We write $\mathrm{End}(A)$ for the integral endomorphism ring of A, and $D_A := \mathrm{End}(A) \otimes_{\mathbb{Z}} \mathbb{Q}$. The algebra D_A is central simple over $\mathbb{Q}$.

For $n \in \mathbb{Z}_{>1}$, we will denote by $P^+(n)$ the largest prime divisor of n. By convention, $P^+(1) = 1$. For $X, B > 0$, write $\Psi(X, B) := \#\{1 \leq n \leq X \mid P^+(n) \leq B\}$ for the number of B-smooth integers $\leq X$, and $\delta(X, B) := \Psi(X, B)/X$ for the probability that a uniformly random integer $n \in \{1, \ldots, X\}$ is B-smooth.

Efficient computation of isogenies

For algorithmic purposes, we will require our abelian surfaces to be given by a sufficiently explicit computational model. For simplicity, and unless mentioned otherwise, we will implicitly assume a level-4 symmetric theta model. This is a

standard assumption in the literature when it comes to algorithmic results related to abelian varieties (cf. [LR15,SPW26]). Equivalently, through polynomial-time reductions, one can represent A as either the canonically polarized Jacobian of an explicit genus-two curve or a product of elliptic curves with the product polarization; see [CR15, §5].

For $p \nmid n$, maximal (n-)isotropic subgroups $H \subseteq A[n]$ are in bijection with polarized isogenies φ with domain A (where $\ker \varphi = H$), up to post-composition by an isomorphism. We call a maximal n-isotropic subgroup isomorphic to $(\mathbb{Z}/n\mathbb{Z})^2$ an (n,n)-*subgroup*, and refer to the corresponding isogeny as an (n,n)-*isogeny*.

We say an *effective representation* of a polarized isogeny $\varphi : A \to B$ of multiplier d is an algorithm that allows to evaluate φ on geometric points $P \in A(\overline{\mathbb{F}_p})$. We call the representation *efficient* if the evaluation of $P \in A(\mathbb{F}_{p^{2r}})$ takes time polynomial in r, $\log p$, and $\log d$.

Theorem 2.1. *Let $A/\mathbb{F}_{p^2}$ be a maximal superspecial ppas. Let $\ell \neq p$ be a prime number. Then there exists a randomized algorithm with expected running time polynomial in ℓ and $\log p$ that enumerates all maximal isotropic subgroups $H \subseteq A[\ell]$, together with, for each such H:*

- **(i)** *the quotient principally polarized abelian surface $C \cong A/H$;*
- **(ii)** *an efficient representation of an (ℓ,ℓ)-isogeny $\varphi : A \to C$.*

Moreover, given bases for two such subgroups H_1, H_2 on A, we can compute a basis for $H_1 \cap H_2$ in time polynomial in ℓ and $\log p$.

Proof. The geometric points of $A[\ell]$ are defined over a field extension of degree polynomial in ℓ (e.g. by [Rob25, Lemma A.3]), hence we can compute a basis for $A[\ell]$ in expected time polynomial in ℓ and $\log p$ [LR12, §5.3]. We can convert this into a symplectic basis using Weil pairings [LR12, Algorithm 6.3]. One can then enumerate all maximal isotropic subgroups and codomains using [LR12, Algorithm 5.7]. Finally, one can obtain efficient representations of the corresponding isogenies through isogeny interpolation [Rob25, §5.3]. One can also use [SPW26, Thm. 6.25, Thm. 7.5, Cor. 7.7] combined with Axioms A16, A17, and A26 from that same paper to obtain the same sequence of results. Finally, computing a basis of the intersection of two subgroups H_1, H_2 can be done using Weil pairings [LR12, Algorithm 6.3].

Quaternionic coordinates

We recall the setting of IKO coordinates [IKO86] as used in [Pha26]. Let (A, λ_A) be a maximal superspecial ppas over $\mathbb{F}_{p^2}$ and let $E/\mathbb{F}_{p^2}$ be a supersingular elliptic curve with endomorphism ring $\mathcal{O} \subseteq B_{p,\infty}$, where $B_{p,\infty}$ denotes the unique quaternion algebra over $\mathbb{Q}$ ramified at p and ∞. We choose an $\mathbb{F}_{p^2}$-isomorphism of unpolarized varieties $\iota_A : A \to E^2$ and put

$$\kappa_A : \mathrm{End}(A) \to \mathrm{M}_2(\mathcal{O}), \quad u \mapsto \iota_A \circ u \circ \iota_A^{-1}.$$

Then, transporting the polarization λ_A to E^2 identifies it with the matrix

$$h = \lambda_0^{-1} \circ (\iota_A^{-1})^\vee \circ \lambda_A \circ \iota_A^{-1} \in \mathrm{M}_2(\mathcal{O}),$$

where λ_0 is the product principal polarization of E^2.

We recall (see [Pha26, §2]) the following facts from the IKO correspondence [IKO86].

(1) κ_A is an isomorphism of rings and the reduced traces coincide.

(2) The matrix h is hermitian, positive definite, with hermitian determinant one:

$$h = \begin{pmatrix} s & r \\ \bar{r} & t \end{pmatrix}, \quad s, t \in \mathbb{Z}_{>0}, \quad r \in \mathcal{O}, \quad st - \mathrm{nrd}(r) = 1.$$

(3) The Rosati involution is transported to $\kappa_A(x^\dagger) = h^{-1}\kappa_A(x)^* h$ for $x \in \mathrm{End}(A)$.

(4) κ_A maps $\mathscr{S}_A$ onto

$$S_h = \{X \in \mathrm{M}_2(\mathcal{O}) \mid X^* h = hX,\ \mathrm{Trd}(X) = 0\},$$

and for $X \in S_h$, we have $X^2 = q(X) I_2$ where $q(X) = \frac{1}{4}\mathrm{Trd}(X^2) \in \mathbb{Z}$.

3 Small degree endomorphisms and the endomorphism ring

Let (A, λ_A) be a maximal superspecial ppas over $\mathbb{F}_{p^2}$. Recall that the Rosati involution is defined on endomorphisms $x \in \mathrm{End}(A)$ by $x^\dagger = \lambda_A^{-1} \circ x^\vee \circ \lambda_A$. As in [Pha26], consider the *Rosati lattice* $\mathscr{S}_A$ defined by

$$\mathscr{S}_A = \{x \in \mathrm{End}(A) \mid x^\dagger = x,\ \mathrm{Trd}_{D_A/\mathbb{Q}}(x) = 0\}, \tag{1}$$

together with the following quadratic form:

$$q_A(x) = \frac{1}{4}\mathrm{Trd}_{D_A/\mathbb{Q}}(x^2). \tag{2}$$

Let us recall some important basic facts about the quadratic module $(\mathscr{S}_A, q_A)$. From [Pha26, Propositions 2.5 and 2.7], we have:

Proposition 3.1. *The lattice $(\mathscr{S}_A, q_A)$ is positive definite, integral, and has rank five. Moreover, we have*

$$x^2 = [q_A(x)]_A \quad (x \in \mathscr{S}_A), \quad \det(\mathscr{S}_A, q_A) = \frac{p^2}{16}. \tag{3}$$

As an immediate corollary of the results proved in [Pha26], we obtain the following:

Proposition 3.2. *A shortest nonzero vector α of $(\mathscr{S}_A, q_A)$ is primitive in* $\mathrm{End}(A)$*, non scalar, and satisfies $\alpha^\dagger\alpha = [N]$ for an integer $1 \leq N \leq (p^2/2)^{1/5}$.*

Proof. This follows from [Pha26, Proposition 4.1] with $C = A$, $\psi = \mathrm{id}_A$, so that $M = 1$. Then, $\mathscr{S}_A = L_\psi$, and the proof of [Pha26, Proposition 4.1] shows precisely that a shortest nonzero vector α satisfies $\alpha^\dagger\alpha = [N]$, with $N \le (p^2/2)^{1/5}$. Since $\mathrm{Trd}(\alpha) = 0$, it cannot be scalar, for $\mathrm{Trd}(a\,\mathrm{id}_A) = 4a$. Finally, for primitivity, if $\alpha = [d]\,\gamma$ with $d > 1$ and γ integral, then since $\mathrm{End}(A)$ is torsion-free, we get $\gamma^\dagger = \gamma$ and $\mathrm{Trd}(\gamma) = 0$. But then, $q_A(\gamma) = N/d^2$, contradicting minimality of α.

The proposition implies that α is a polarized isogeny of multiplier N. However, we can say more:

Theorem 3.3. *Let $\alpha \in \mathscr{S}_A$ be primitive in $\mathrm{End}(A)$, and assume that $N = q_A(\alpha) > 0$ and coprime to p. Then α is an (N, N)-isogeny. In particular, a shortest nonzero vector $\alpha \in \mathscr{S}_A$ is an (N, N)-isogeny, where $N \le \lfloor (p^2/2)^{1/5} \rfloor$. More generally, let α_i, $1 \le i \le 5$, be linearly independent vectors attaining the successive minima of $(\mathscr{S}_A, q_A)$, and put $N_i = q_A(\alpha_i)$ for each i. Then, each α_i is primitive in $\mathrm{End}(A)$, and whenever $p \nmid N_i$, α_i is an (N_i, N_i)-isogeny.*

We first prove two basic lemmas.

Lemma 3.4. *Let $N \ge 1$ be an integer, and let V be a finite abelian group with $\#V = N^4$, and $NV = 0$, i.e., $V[N] = V$. Let $a : V \to V$ be a homomorphism with $a \circ a = 0$, and let $K = \ker a$. If for every prime $\ell \mid N$ we have $\dim_{\mathbb{F}_\ell} K[\ell] \le 2$, then $a(V) = K$ and $K \cong (\mathbb{Z}/N\mathbb{Z})^2$.*

Proof. From $a \circ a = 0$, we have $a(V) \subset K$. Since $V/K \cong a(V)$, we have $|a(V)| = |V|/|K|$, so $N^4 = |V| = |K| \cdot |a(V)| \le |K|^2$, i.e., $N^2 \le |K|$. In particular $K \neq 0$ if $N \ge 2$. For $N = 1$ there is nothing to prove. As a finite abelian group, $K \cong \bigoplus_{i=1}^{r} \mathbb{Z}/d_i\mathbb{Z}$ for some $r \ge 1$ and positive integers d_i with $d_i \mid d_{i+1}$, $1 \le i \le r-1$. Since $NK = 0$, each $d_i \mid N$, and any prime dividing d_1 divides every d_i and divides N. Let ℓ be such a prime. Since ℓ-torsion commutes with isomorphisms and direct sums, we have $K[\ell] \cong \bigoplus_{i \ge 1} (\mathbb{Z}/d_i\mathbb{Z})[\ell]$.

For any positive integer d divisible by ℓ, put $M = \mathbb{Z}/d\mathbb{Z}$ and consider the homomorphism

$$\varphi : \mathbb{Z} \to \mathbb{Z}/d\mathbb{Z} = M, \quad x \mapsto (d/\ell)x + d\mathbb{Z}.$$

For $y \in \mathbb{Z}$, we have $y \in M[\ell]$ if and only if $\ell y \in d\mathbb{Z}$, if and only if $y \in (d/\ell)\mathbb{Z}$, if and only if $y = (d/\ell)x$ for some $x \in \mathbb{Z}$, i.e., $y \in \varphi(\mathbb{Z})$. Thus, $\varphi(\mathbb{Z}) = M[\ell]$. On the other hand, $x \in \ker\varphi$ if and only if $(d/\ell)x \in d\mathbb{Z}$, if and only if $x/\ell \in \mathbb{Z}$, i.e., $x \in \ell\mathbb{Z}$. Thus, for any i, we have $(\mathbb{Z}/d_i\mathbb{Z})[\ell] \cong \mathbb{Z}/\ell\mathbb{Z}$, and hence $K[\ell] \cong (\mathbb{Z}/\ell\mathbb{Z})^r$.

The assumption means that $r = \dim_{\mathbb{F}_\ell} K[\ell] \le 2$, whence

$$N^2 \le |K| = d_1 \cdots d_r \le N^r \le N^2.$$

Thus, $r = 2$, and $d_1 d_2 = N^2$. Since $d_1, d_2 \le N$, we obtain $d_1 = d_2 = N$, $|K| = N^2$, so $|a(V)| = N^4/N^2 = N^2 = |K|$. Since $a(V) \subset K$, we obtain $a(V) = K$. □

Recall that we have an isomorphism $\kappa_A : \mathrm{End}(A) \xrightarrow{\sim} \mathrm{M}_2(\mathcal{O})$ with

$$\kappa_A(\mathscr{S}_A) = S_h = \{X \in \mathrm{M}_2(\mathcal{O}) \mid X^* h = hX,\ \mathrm{Trd}(X) = 0\}.$$

Following [Pha26, Lemma 3.4], we also have an isomorphism $\rho : \mathrm{M}_2(\mathcal{O}_\ell) \xrightarrow{\sim} \mathrm{M}_4(\mathbb{Z}_\ell)$ induced by the isomorphism $\iota : \mathcal{O} \otimes \mathbb{Z}_\ell \xrightarrow{\sim} \mathrm{M}_2(\mathbb{Z}_\ell)$. We have

$$\iota(\bar{b}) = J_2^{-1}[\iota(b)]^T J_2, \quad J_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

and, with $\Omega = J_0 \rho(h)$ with $J_0 = \mathrm{diag}(J_2, J_2)$, ρ maps $S_h \otimes \mathbb{Z}_\ell$ onto

$$S_\Omega = \{T \in \mathrm{M}_4(\mathbb{Z}_\ell) \mid \Omega T = T^T \Omega,\ \mathrm{tr}(T) = 0\}.$$

(See the proof of [Pha26, Proposition 3.6].) There is $U \in \mathrm{GL}_4(\mathbb{Z}_\ell)$ such that $U^T \Omega U = J$, where $J = \left(\begin{smallmatrix} 0 & I_2 \\ -I_2 & 0 \end{smallmatrix}\right)$. The conjugation map $T \mapsto U^{-1} T U$ then maps S_Ω onto

$$S_J = \{Y \in \mathrm{M}_4(\mathbb{Z}_\ell) \mid Y^T J = JY,\ \mathrm{tr}(Y) = 0\}.$$

The elements of S_J are elements of the form

$$Y(z,x,y,d,e) = \left(\begin{array}{cc|cc} z & x & 0 & d \\ y & -z & -d & 0 \\ \hline 0 & e & z & y \\ -e & 0 & x & -z \end{array}\right) = \begin{pmatrix} P & dJ_2 \\ eJ_2 & P^T \end{pmatrix},$$

with $z, x, y, d, e \in \mathbb{Z}_\ell$ (see [Pha26, (3.10)]).

Lemma 3.5. *Let $\ell \neq p$ be a prime, and let $X \in S_h$ with $X \notin \ell\,\mathrm{M}_2(\mathcal{O})$. Then, for $T = \rho(X)$, we have* $\mathrm{rank}_{\mathbb{F}_\ell} \overline{T} \geq 2$.

Proof. By what we recalled above, we have $U^{-1}TU = Y(z,x,y,d,e)$. Since $\mathrm{M}_2(\mathcal{O}) \cap \ell\,\mathrm{M}_2(\mathcal{O}_\ell) = \ell\,\mathrm{M}_2(\mathcal{O})$, we have $X \notin \ell\,\mathrm{M}_2(\mathcal{O}_\ell)$, hence $T \notin \ell\,\mathrm{M}_4(\mathbb{Z}_\ell)$. Thus, $U^{-1}TU \notin \ell\,\mathrm{M}_4(\mathbb{Z}_\ell)$ as well. Denote by $\bar{Y}$ the reduction modulo ℓ of Y. Reduction modulo ℓ then implies that $\bar{Y} = \bar{U}^{-1}\bar{T}\bar{U}$ is nonzero, with the same rank as $\bar{T}$. If $\bar{d} \neq 0$ or $\bar{e} \neq 0$ looking at the block matrix shows that we have at least two linearly independent rows. If $\bar{d} = \bar{e} = 0$, then $\bar{Y} = \mathrm{diag}(\bar{P}, \bar{P}^T)$ with $\bar{P} \neq 0$ since $\bar{Y} \neq 0$, so $\mathrm{rank}(\bar{T}) = \mathrm{rank}(\bar{Y}) = 2 \cdot \mathrm{rank}(\bar{P}) \geq 2$. □

We can now prove Theorem 3.3.

Proof (of Theorem 3.3). We already know that $\alpha^\dagger \alpha = [N]$ and $\deg \alpha = N^2$. Since $p \nmid N$, we need to prove $\ker \alpha(\bar{k}) \cong (\mathbb{Z}/N\mathbb{Z})^2$.

Let $V = A[N](\bar{k})$. Then, $|V| = N^4$ and $NV = 0$. Since α commutes with $[N]_A$, it maps V to V, and we denote the restriction by $a = \alpha|_V : V \to V$. Since $\alpha^2 = [N]_A$, $a \circ a = N \cdot \mathrm{id}_V = 0$, and $a(V) = \alpha(A[N](\bar{k}))$, so $\ker a = \ker \alpha(\bar{k})$ since V contains $\ker \alpha(\bar{k})$.

If we can show that $(\ker a)[\ell]$ has $\mathbb{F}_\ell$-dimension ≤ 2 for every prime divisor ℓ of N, Lemma 3.4 then implies the desired result. Let $\ell \mid N$ be a prime. Let

$W = A[\ell](\bar{k}) = A(\bar{k})[\ell]$. This is an $\mathbb{F}_\ell$-vector space of dimension 4. We have $W \subset V$ since $\ell \mid N$, and since $\ell\alpha(P) = \alpha(\ell P)$, we have $\alpha(W) \subset W$, so α maps W into itself. Therefore,

$$\begin{aligned}(\ker a)[\ell] &= \{P \in V \mid \alpha(P) = 0,\, \ell P = 0\} \\ &= \{P \in W \mid \alpha(P) = 0\} \\ &= \ker(\alpha|_W).\end{aligned}$$

It suffices to show that

$$\operatorname{rank}_{\mathbb{F}_\ell}(\alpha|_W) \geq 2.$$

First, observe that $\alpha \notin \ell\,\mathrm{End}(A)$, which follows from primitivity: if $\alpha = \ell\beta$ with $\beta \in \mathrm{End}(A)$, then $\ell\beta^\dagger = \alpha^\dagger = \alpha = \ell\beta$, hence $\beta^\dagger = \beta$ since $\mathrm{End}(A)$ is torsion-free. Since also $\mathrm{Trd}(\beta) = 0$, this would contradict primitivity. Hence, $X = \kappa_A(\alpha) \in S_h \setminus \ell\,\mathrm{M}_2(\mathcal{O})$, so $T = \rho(X)$ satisfies the conditions of Lemma 3.5, and $\operatorname{rank}_{\mathbb{F}_\ell}(\bar{T}) \geq 2$.

To conclude, we need to relate $\alpha|_W$ and $\bar{T}$. Put $R = \mathrm{End}(A)$ to lighten notation. Note that a priori, we do not know that the matrix $\bar{T}$ (with its standard action) is related to the action of α on torsion points. We now show that the action of $R/\ell R$ on W is, up to a change of basis, precisely the standard representation of the matrix ring $\mathrm{M}_4(\mathbb{F}_\ell) \cong R/\ell R$. Our identification of $R/\ell R$ with the corresponding $\mathbb{Z}_\ell$ scalar extension implies that the ring isomorphism $\rho \circ (\kappa \otimes \mathbb{Z}_\ell) : R \otimes \mathbb{Z}_\ell \to \mathrm{M}_4(\mathbb{Z}_\ell)$ induces a ring isomorphism

$$\varphi : R/\ell R \xrightarrow{\sim} \mathrm{M}_4(\mathbb{F}_\ell), \quad x + \ell R \mapsto \overline{\rho(\kappa(x))},$$

with $\varphi(\alpha + \ell R) = \bar{T}$. Each $x \in R$ maps W into itself, and the restriction to W,

$$r : \mathrm{End}(A) \to \mathrm{End}_{\mathbb{F}_\ell}(W), \quad x \mapsto x|_W,$$

is a homomorphism of rings, whose kernel contains ℓR since $(\ell y)|_W = \ell \cdot (y|_W) = 0$, so it factors through the quotient $\pi : R \to R/\ell R$. There is a unique homomorphism of rings $\bar{r} : R/\ell R \to \mathrm{End}_{\mathbb{F}_\ell}(W)$ such that $r = \bar{r} \circ \pi$. Put $\theta = \bar{r} \circ \varphi^{-1}$. Then, we have the following commutative diagram

$$\begin{array}{ccc} R & \xrightarrow{\;r\;} & \mathrm{End}_{\mathbb{F}_\ell}(W) \\ {\scriptstyle\pi}\downarrow & \nearrow_{\bar{r}} & \uparrow{\scriptstyle\theta} \\ R/\ell R & \xrightarrow[\;\varphi\;]{} & \mathrm{M}_4(\mathbb{F}_\ell) \end{array}$$

and θ is a homomorphism of rings, with $\theta(\bar{T}) = \bar{r}(\alpha + \ell R) = \alpha|_W$. Let E_{ij} denote the matrix units of $\mathrm{M}_4(\mathbb{F}_\ell)$, so that we can decompose $W = \bigoplus_i \theta(E_{ii})W$, with $\theta(E_{ij})\theta(E_{jj})W = \theta(E_{ii})W$. So the four subspaces $\theta(E_{ii})W$ are one-dimensional. Pick any nonzero $w_1 \in \theta(E_{11})W$ and put $w_i = \theta(E_{i1})w_1 \in \theta(E_{ii})W$ for the others. The vectors w_i's form a basis of W in which $Y \in \mathrm{M}_4(\mathbb{F}_\ell)$ is the matrix of $\theta(Y)$. In particular, $\theta(\bar{T}) = \alpha|_W$ has matrix $\bar{T}$. We transfer the lower bound on the rank from $\bar{T}$ to $\alpha|_W$, which completes the proof for the shortest vector.

Let μ_i, $1 \leq i \leq 5$, denote the successive minima. The same primitivity argument as for the shortest vector holds. Indeed, fix some i. If $\alpha_i = d\beta_i$ with β_i integral and $d \geq 2$, since $\mathrm{End}(A)$ is torsion-free, this implies $\beta_i \in \mathscr{S}_A$ and $q_A(\beta_i) = \mu_i/d^2 < \mu_i$. Replacing α_i by β_i preserves linear independence yet decreases the value of $q_A(\beta_i)$, a contradiction. This completes the proof. □

4 On the number of maximal n-isotropic subgroups of an abelian surface

For this section, let k denote an algebraically closed field of characteristic $p > 0$. For any ppas A/k, we will give bounds on the total number of outgoing polarized isogenies (up to post-composition with an isomorphism) of multiplier n, i.e. on the total number of maximal n-isotropic subgroups, summed over all n up to some bound $X < p$. This will be relevant when bounding the cost of our main algorithms, which will enumerate (a subset of) all outgoing polarized isogenies of multiplier $\leq X$. The results in this section also extend to the case where $\mathrm{char}(k) = 0$, without any restriction on X.

4.1 Warm up: dimension one

If E/k is an elliptic curve, then the number $a_1(n)$ of cyclic subgroups of order n, for $p \nmid n$, is equal to $\psi(n)$, where ψ is the Dedekind-ψ function, defined as the unique multiplicative function $\mathbb{Z} \to \mathbb{Z}$ satisfying $\psi(\ell^e) = (\ell+1)\ell^{e-1}$ for every prime ℓ. The number $b_1(n)$ of (not necessarily cyclic) subgroups of order n is given by the sum-of-divisors function $b_1(n) = \sigma_1(n) = \sum_{d|n} d$. We clearly have $a_1(n) \leq b_1(n)$.

Lemma 4.1. *Let $X \in \mathbb{Z}_{>0}$. Then $\sum_{n \leq X} \psi(n) \leq \sum_{n \leq X} \sigma_1(n) \leq X^2$.*

Proof. The first inequality follows from the fact that $a_1(n) \leq b_1(n)$. For the second, note that

$$\sum_{n \leq X} \sigma_1(n) = \sum_{n \leq X} \sum_{d|n} d = \sum_{d=1}^{X} d\lfloor X/d \rfloor \leq \sum_{d=1}^{X} X = X^2.$$

That is, if $X < p$, then the total number of isogenies with domain E and of degree $\leq X$, up to post-composition by an isomorphism, is uniformly bounded by X^2.

4.2 Dimension two

Let A/k be a principally polarized abelian surface. We will denote the two-dimensional analogues of the counting functions discussed above by a_2 and b_2, which count respectively: the number of maximal isotropic subgroups isomorphic to $(\mathbb{Z}/n\mathbb{Z})^2$ (corresponding to (n,n)-isogenies), and the total number of maximal

isotropic subgroups of $A[n]$ (corresponding to polarized isogenies with multiplier n).

It is well known that for every prime number $\ell \neq p$ we have $a_2(\ell) = b_2(\ell) = (\ell^2+1)(\ell+1)$, and that, given a fixed (ℓ,ℓ)-isogeny φ_1, the total number of *good extensions* φ_2, i.e. those (ℓ,ℓ)-isogenies φ_2 for which the composition $\varphi_2 \circ \varphi_1$ is an (ℓ^2,ℓ^2)-isogeny, is ℓ^3; see for example [LS02, Prop. 4.4]. Therefore, we obtain, for prime powers ℓ^e and $n \in \mathbb{Z}_{>0}$ respectively,

$$a_2(\ell^e) = \ell^{3(e-1)}(\ell^2+1)(\ell+1); \qquad a_2(n) = n^3 \prod_{\ell \mid n}(1+\ell^{-1})(1+\ell^{-2}).$$

Counting the total number of maximal n-isotropic subgroups is slightly more involved, but a closed expression can also be found in the literature.

Theorem 4.2. *We have* $b_2(n) = \sum_{k \mid n} k^3 \sigma_1(n^2/k^2)$.

Proof. This is the case $d_1 = d_2 = n$ in [BOPY18, Thm. 3.2].

We will now prove a uniform upper bound for $\sum_{n \leq X} b_2(n)$, i.e. the total number of polarized isogenies with domain A of multiplier $\leq X$, where $X < p$. First, we record the following lemma.

Lemma 4.3. *Let* $X \in \mathbb{Z}_{>0}$. *Then* $\sum_{n \leq X} \sigma_1(n^2) \leq X^3$.

Proof. We have $\sigma_1(n^2) = \sum_{d \mid n} d\psi(d)$ by e.g. [Mat12, Eq. (3.89)&(3.93)]. Now

$$\sum_{n \leq X} \sigma_1(n^2) = \sum_{d \leq X} d\psi(d)\lfloor X/d \rfloor \leq X \sum_{d \leq X} \psi(d) \leq X^3,$$

where we used Lemma 4.1 for the final inequality.

Finally, we prove the desired bound.

Proposition 4.4. *Let* $X \in \mathbb{Z}_{>0}$ *such that* $X < p$. *Then* $\sum_{n \leq X} a_2(n) \leq \sum_{n \leq X} b_2(n) \leq X^4$.

Proof. The first inequality is clear. For the second, note that

$$\sum_{n \leq X} b_2(n) = \sum_{n \leq X} \sum_{k \mid n} k^3 \sigma_1\left(\frac{n^2}{k^2}\right) = \sum_{k \leq X} k^3 \sum_{m \leq X/k} \sigma_1(m^2) \leq \sum_{k \leq X} k^3 \frac{X^3}{k^3} = X^4,$$

where the inequality is by Lemma 4.3.

5 The attack

We now return to the case where the base field of our abelian surfaces is $\mathbb{F}_{p^2}$, where $p > 3$ is a prime number $\equiv 3 \pmod 4$. The goal of this section is to describe an algorithm that, on input of a maximal superspecial principally polarized abelian $A/\mathbb{F}_{p^2}$, returns a non-scalar polarized endomorphism of A (aka a solution to the OneEnd problem). Under a heuristic assumption about the smoothness probability of the multiplier of one special such endomorphism, we show (Algorithm 2) that this can be done in expected time and memory $p^{4/5+o(1)}$.

5.1 The heuristic

Our main heuristic is motivated by the dimension-two equivalent of the standard principle that the degree of a *random* isogeny is B-smooth with roughly the same probability as a random integer of the same size; this is the same motivating principle driving the $p^{1/3+o(1)}$-attack against the one-dimensional isogeny problem, cf. [Wes26, Heuristic 1].

We will only require a B-smoothness prediction for B equal to a certain explicit subexponential function of X, and we allow an error in the smoothness probability up to a factor of $B^{o(1)}$. Our heuristic is motivated by the following classical result.

Recall that we denote by $\delta(X, B) = \Psi(X, B)/X$ the probability that a uniformly random integer $1 \leq n \leq X$ is B-smooth.

Theorem 5.1. *Let $X \in \mathbb{Z}_{>0}$ and let $B = \exp\left(\frac{1}{\sqrt{2}}\sqrt{\log X \log\log X}\right)$. Then $\delta(X, B) = B^{-1+o(1)}$ as $X \to \infty$.*

Proof. This follows immediately from classical asymptotics for the Dickman–de Bruijn function; e.g. [HT93, Cor. 1.3].

Now, recall that, by Theorem 3.3, every maximal superspecial principally polarized abelian surface $A/\mathbb{F}_{p^2}$ contains an (n, n)-endomorphism $\alpha \in \mathrm{End}(A)$, where $n \leq \lfloor (p^2/2)^{1/5} \rfloor$. The following is our main heuristic assumption.

Heuristic 1 *Let p be a prime number, and let $A/\mathbb{F}_{p^2}$ be a uniformly random maximal superspecial principally polarized abelian surface. Let $X = \lfloor (p^2/2)^{1/5} \rfloor$, and $B = \exp\left(\frac{1}{\sqrt{2}}\sqrt{\log X \log\log X}\right)$. Let $\alpha \in \mathscr{S}_A \subseteq \mathrm{End}(A)$ be a shortest nonzero vector, with corresponding multiplier $n \leq X$. Then, in the limit $p \to \infty$, we have that n is B-smooth with probability at least $B^{-1+o(1)}$.*

In Section 9, we will provide experimental evidence for this heuristic, and compare the B-smoothness probability of the multiplier n of α against what one would expect if n were sampled randomly from $\{1, \ldots, X\}$, where we consider various possible probability distributions on this set.

Remark 5.2. It will be apparent that Heuristic 1 can be slightly weakened for the purpose of solving OneEnd (Algorithm 2 and Theorem 5.4). Namely, we only need to assume the existence of *one* non-scalar (n, n)-endomorphism, where $n \leq X$, such that n is B-smooth with probability at least $B^{-1+o(1)}$; in particular, the statement of Heuristic 1 can be made independent of $\alpha \in \mathscr{S}_A$. In Section 6, however, we will require B-smoothness of the multiplier of α specifically (i.e. the current Heuristic 1). The experimental results in Section 9 also support this stronger heuristic.

5.2 The algorithms

Let $\ell \neq p$ be a prime number. Following Theorem 2.1, we denote by $\mathsf{EnumerateIsotropicSubgroups}_\ell(\cdot)$ an algorithm that takes as input a maximal superspecial ppas $A/\mathbb{F}_{p^2}$ and outputs all maximal isotropic subgroups $H \subseteq A[\ell]$. We denote by $\mathsf{EfficientIsogeny}_\ell(\cdot)$ an algorithm that takes as input a maximal ℓ-isotropic subgroup H, and outputs a pair (C, φ), where $C \cong A/H$ and $\varphi : A \to C$ is an efficient representation of an (ℓ, ℓ)-isogeny with kernel H. On input of such an isogeny φ, we denote by $\mathsf{DualKernel}_\ell(\cdot)$ an algorithm that computes the (ℓ, ℓ)-subgroup $\hat{H} \subseteq C[\ell]$ which is the kernel of the adjoint isogeny of φ. The cost of $\mathsf{DualKernel}_\ell(\cdot)$ is polynomial in $\ell + \log p$: indeed, one can obtain a generating set of the correct (ℓ, ℓ)-subgroup by evaluating φ at a basis of $A[\ell]$. Note that one can therefore also compute the adjoint isogeny in time $(\ell + \log p)^{O(1)}$.

The main algorithm fueling our method for OneEnd is Algorithm 1.

Algorithm 1 Enumerating smooth (n, n)-isogenies.

Require: A maximal superspecial ppas $A_0/\mathbb{F}_{p^2}$, a smoothness bound $B \in \mathbb{Z}_{>0}$, and a multiplier bound $X \in \mathbb{Z}_{>0}$, such that $B \leq X < p$.

Ensure: A set L containing, for each B-smooth integer $1 \leq n \leq X$ and each (n, n)-subgroup $H \subseteq A_0[n]$, a tuple (A, φ, n), where $A \cong A_0/H$ and $\varphi : A_0 \to A$ is an efficient representation of an (n, n)-isogeny with kernel H.

```
L ← {(A_0, id_{A_0}, 1)}
for all primes ℓ ≤ B do
    T ← {(A, φ, n, {0_A}) | (A, φ, n) ∈ L}
    for e = 0, ..., ⌊log_ℓ(X)⌋ do
        T_e ← T
        for all (A, φ, n, Ĥ) ∈ T_e such that ℓ^e ∥ n and ℓ · n ≤ X do
            for all H ∈ EnumerateIsotropicSubgroups_ℓ(A) with H ∩ Ĥ = {0_A} do
                C, ψ ← EfficientIsogeny_ℓ(H)
                D̂ ← DualKernel_ℓ(ψ)
                T ← T ∪ {(C, ψ ∘ φ, ℓ · n, D̂)}
            end for
        end for
    end for
    L ← {(A, φ, n) | (A, φ, n, Ĥ) ∈ T}
end for
return L
```

Theorem 5.3. *Algorithm 1 is correct and uses expected time and memory* $X^4(B + \log p)^{O(1)}$.

Proof. Every (n, n)-isogeny $A_0 \to A'$ can be decomposed into a chain of isogenies η_i of prime multiplier, with the multiset $\mathcal{M}$ of prime multipliers determined by the factorization of n, in such a way that every η_i is a good extension of η_{i-1},

i.e. $\ker \eta_i \cap \ker \eta_{i-1}^\dagger = \{0\}$ (cf. [CDS20, Def. 2]). Moreover, up to intermediate isomorphisms, there is exactly one such chain for every ordering on $\mathcal{M}$. The algorithm enumerates, for every n, all chains of good extensions of total multiplier n, by choosing a fixed ordering on the prime degrees. It therefore obtains an element (A, φ) of L for every (n, n)-subgroup of A_0.

The complexity of the algorithm is dominated by the cost of populating T. Each new entry of T costs at most one evaluation of each of the subroutines $\mathsf{EnumerateIsotropicSubgroups}_\ell$, $\mathsf{EfficientIsogeny}_\ell$, $\mathsf{DualKernel}_\ell$, and at most two (ℓ, ℓ)-subgroup intersections $H \cap \hat{H}$.[2], and each of these subroutines takes expected time polynomial in $\ell + \log p \leq B + \log p$, by Theorem 2.1.

The number of entries of T equals the cardinality of the output set L, which is bounded above by $\sum_{n \leq X} a_2(n) \leq X^4$ by Proposition 4.4. Therefore, the total expected running time of the algorithm is $X^4(B + \log p)^{O(1)}$.

The memory is dominated by the total size of T (the other lists add at most a constant factor of overhead). Let $(A, \varphi, n, \hat{H})$ be an entry of T. Since $B \leq X < p$, we directly see that the bitsize of A, n, and $\hat{H}$ can be bounded by $(B + \log p)^{O(1)}$. Moreover, we know that φ is a composition of at most $\log_2(X)$ efficient representations of isogenies of multiplier $\leq B$. It follows that the total size of φ is polynomial in $\log_2(X)(\log B + \log p) = (\log p)^{O(1)}$. We conclude that the total memory requirement is indeed $X^4(B + \log p)^{O(1)}$.

Now, write $\mathsf{RandomWalk}(\cdot)$ for a routine that takes a random walk in the $(2, 2)$-isogeny graph from an input superspecial ppas A, and returns the endpoint C, together with an efficient representation of the isogeny $\psi : A \to C$. Here, we assume the random walk is of sufficient length to approximate the uniform distribution (or, more precisely, the stationary distribution, which is weighted by the inverse of the size of the automorphism group), with error, say, $< p^{-3}$; this is possible in $O(\log p)$ steps by rapid-mixing properties of the $(2, 2)$-isogeny graph [ATY24,SKTW26].

We also write $\mathsf{ComputePolarizedIsomorphism}(C_1, C_2)$ for a routine that computes an isomorphism between a pair of isomorphic ppass C_1, C_2. This can be done in time $O(\log p)$ by [SPW26, Prop. 6.19 + Cor. 6.20].

Theorem 5.4. *Assume Heuristic 1. Algorithm 2 is correct and runs in expected time and memory $p^{4/5+o(1)}$.*

Proof. The output of the algorithm is indeed a non-integer polarized endomorphism, because, by construction, the two distinct entries of L correspond to isogenies φ_1, φ_2 with distinct kernels.

The cost of one call to Algorithm 1, regarding both time and memory, is $(X^{1/2}B^{1/2})^4(B+\log p)^{O(1)}$ by Theorem 5.3. Since $B = X^{o(1)} = p^{o(1)}$ and $X^{1/2} < p^{1/5}$, this is $p^{4/5+o(1)}$. Moreover, we can check if L contains distinct entries

[2] Indeed, if $H \cap \hat{H} \neq \{0\}$, then no new entry to T is produced. However, since the total number of maximal ℓ-isotropic subgroups H is $(\ell^2+1)(\ell+1) \leq 2 \cdot \ell^3$, where ℓ^3 is a lower bound on the number of good extensions, we need to check at most two subgroups H for every new entry to T.

Algorithm 2 Find a nontrivial polarized endomorphism.

Require: A maximal superspecial ppas $A/\mathbb{F}_{p^2}$.
Ensure: A polarized endomorphism $\beta \in \mathrm{End}(A) \setminus \mathbb{Z}$.

```
X ← ⌊(p²/2)^{1/5}⌋
B ← exp((1/√2)·√(log X log log X))
while true do
    C, ψ ← RandomWalk(A)
    L ← the list returned by Algorithm 1 on input (C, B, X^{1/2}B^{1/2}).
    if L contains two entries (C_1, φ_1, n_1), (C_2, φ_2, n_2) for which C_1 ≅ C_2 then
        u ← ComputePolarizedIsomorphism(C_1, C_2)
        break
    end if
end while
return ψ^† ∘ φ_2^† ∘ u ∘ φ_1 ∘ ψ
```

$(C_1, \varphi_1, n_1), (C_2, \varphi_2, n_2)$ for which $C_1 \cong C_2$ in time and memory $\#L(\log \#L + \log p)^{O(1)}$: for example, one can compute the absolute Igusa invariants of every entry, sort the list, and then compare consecutive entries.[3] Since the total number of entries of L is $\leq X^2B^2$, this can also be done in time and memory $p^{4/5+o(1)}$.

By Heuristic 1 (in fact, the weaker version given by Remark 5.2), the probability that an (n, n)-endomorphism γ of C exists, where $n \leq X$ is B-smooth, is at least $B^{-1+o(1)}$. Since such an (n, n)-endomorphism γ can always be written as the composition of two (m_i, m_i)-isogenies, with $m_i \leq X^{1/2}B^{1/2}$ $(i \in \{1, 2\})$, the existence of γ guarantees a collision in L (i.e. two distinct entries corresponding to isomorphic ppass), which will thus be detected by the algorithm. Note that we brushed over one pathological case, which arises when γ can be written as $\varphi^\dagger \circ u \circ \varphi$ for an *automorphism* $u \notin \{\pm 1\}$. However, the number of superspecial ppass admitting a non-scalar polarized automorphism is $\Theta(p^2)$[FS22, Table 2 + §6.2], hence such surfaces occur with density $\sim p^{-1}$. Ignoring this case therefore does not affect the success probability of finding a collision in L asymptotically. The expected number of iterations of the while loop is therefore $B^{1+o(1)}$, yielding a total expected time of $p^{4/5+o(1)}B^{1+o(1)} = p^{4/5+o(1)}$.

6 Obtaining a basis for the Rosati sublattice

Recall that we defined the rank-5 sublattice $\mathscr{S}_A$ of $\mathrm{End}(A)$ as

$$\mathscr{S}_A = \{x \in \mathrm{End}(A) \mid x^\dagger = x,\ \mathrm{Trd}_{D_A/\mathbb{Q}}(x) = 0\}, \tag{4}$$

[3] Alternatively, one can use any tuple of efficiently computable absolute invariants combined with a hash map. Also note that, in the unlikely event that we discover a product surface $C_i \cong E_1 \times E_2$ (for which the Igusa invariants blow up), we can use the pair of j-invariants $(j(E_1), j(E_2))$ instead (or, even better, reduce the problem of finding a non-trivial endomorphism of A directly to the dimension-one case.)

Algorithm 2 finds a non-scalar polarized endomorphism $\gamma = \varphi_2^\dagger \circ u \circ \varphi_1 \in \mathrm{End}(C)$, and pulls it back to obtain $\beta = \psi^\dagger \circ \gamma \circ \psi \in \mathrm{End}(A)$. Since the algorithm stops at the first polarized endomorphism γ found, it need not be that $\gamma \in \mathscr{S}_C$. However, since the adjoint [SPW26, Lemma 6.7] and the trace [SPW26, Prop. 4.10] of efficiently represented endomorphisms can be computed in time $(\log p)^{O(1)}$, we can efficiently verify whether $\gamma \in \mathscr{S}_C$, or, equivalently, $\beta \in \mathscr{S}_A$. Therefore, after a minor adjustment to Algorithm 2, we may ensure that it always returns an element $\beta \in \mathscr{S}_A$.

Algorithm 3 Find an element of $\mathscr{S}_A$.

Require: A maximal superspecial ppas $A/\mathbb{F}_{p^2}$.
Ensure: A polarized endomorphism $\beta \in \mathscr{S}_A \subseteq \mathrm{End}(A) \setminus \mathbb{Z}$.

```
X ← ⌊(p²/2)^{1/5}⌋
B ← exp((1/√2)·√(log X log log X))
while true do
    C, ψ ← RandomWalk(A)
    L ← the list returned by Algorithm 1 on input (C, B, X^{1/2} B^{1/2}).
    if L contains distinct entries (C_1, φ_1, n_1), (C_2, φ_2, n_2) for which C_1 ≅ C_2 then
        u ← ComputePolarizedIsomorphism(C_1, C_2)
        if φ_2^† ∘ u ∘ φ_1 ∈ S_C then
            break
        end if
    end if
end while
return ψ^† ∘ φ_2^† ∘ u ∘ φ_1 ∘ ψ
```

Theorem 6.1. *Assume Heuristic 1. Algorithm 3 is correct and runs in expected time and memory* $p^{4/5+o(1)}$.

Proof. This follows the same proof as Theorem 5.4, now noting that we can ensure, by the argument above, that $\gamma = \varphi_2^\dagger \circ u \circ \varphi_1 \in \mathscr{S}_C$, without affecting the complexity bound.

Now assume the following additional heuristic.

Heuristic 2 *After an expected* $(\log p)^{O(1)}$ *number of applications of Algorithm 3, the outputs* $\beta_i \in \mathscr{S}_A$ *generate a rank-5 sublattice* $L \subseteq \mathscr{S}_A$*, where* $[\mathscr{S}_A : L] = 2^e$ *for some* $e = O(\log p)$.

This is motivated by similar heuristics used in dimension one [EHL+18, Prop. 8] for generating endomorphism rings through pullbacks of random walks. More recently, similar heuristics have been studied in dimension two; cf. [FIK+25, §6.1–6.2, Remark 6.3]. Our assumption is that the special endomorphisms produced by Algorithm 3 generate the rank-5 lattice away from 2. This is an additional assumption on the distribution; see [PW24, §1.2] for obstructions to such heuristics for arbitrary OneEnd oracles.

Proposition 6.2. *Let $\{\beta_i\}$ be a set of generators from Heuristic 2. From it, we can obtain a basis for $\mathscr{S}_A$ in time $(\log p)^{O(1)}$.*

Proof. Denote by L the lattice generated by the β_i. We may assume that the β_i are linearly independent, by invoking standard lattice algorithms that turn a generating set into a basis; cf. [ABI+25, proof of Prop. 5.1]. Since the β_i are efficiently represented and of multiplier bounded by $p^{O(1)}$, we can evaluate them on elements of $A[\ell]$ in time $(\ell + \log p)^{O(1)}$. In particular, by computing the action of β_i on $A[2]$, we can compute, using linear algebra, a linear combination $\sigma = \sum_i b_i \beta_i$ (where $b_i \in \{0, 1\}$, not all zero), such that σ vanishes on $A[2]$, or decide that such a σ does not exist, in which case we are done. Then, using a generic division algorithm based on isogeny interpolation (see [Rob25, Prop. 6.6] for the one-dimensional case, and use [SPW26, Thm. 7.1] to extend to arbitrary dimension), we can compute an efficient representation of $\sigma/2 \in \mathscr{S}_A$. The set $\{\beta_i\} \cup \{\sigma/2\}$ now generates a lattice $L' \supsetneq L$ of smaller index in $\mathscr{S}_A$, and we can again extract a basis $\{\beta_i'\}$ for L'. After at most $e = O(\log p)$ iterations, we obtain a basis for $\mathscr{S}_A$.

7 Finding generators of the endomorphism ring from the Rosati sublattice

We have now produced a basis of the rank-5 lattice $\mathscr{S}_A$. The goal of this section is to show that this is already sufficient to produce a full generating set for $\mathrm{End}(A)$. The key will be the following result.

Theorem 7.1. *For every maximal superspecial ppas (A, λ_A), the lattice of Rosati self-adjoint trace-zero endomorphisms generates $\mathrm{End}(A)$ as a ring. More precisely, let $(e_1, \ldots, e_5)$ be a $\mathbb{Z}$-basis of $\mathscr{S}_A$, and let $X_i \in \mathrm{M}_2(\mathcal{O})$ be the matrix of e_i in IKO coordinates. Then the matrices $X_I = X_{i_1} \cdots X_{i_r}$, $I = \{i_1 < \cdots < i_r\}$, span $\mathrm{M}_2(\mathcal{O})$ over $\mathbb{Z}$.*

The second assertion follows from the first by applying the Poincaré–Birkhoff–Witt theorem for Clifford algebras [Knu91,Bou59]. To prove Theorem 7.1, we will establish a more general result using the integral Clifford algebra of $\mathscr{S}_A$.

7.1 The integral Clifford algebra

Let L be a free $\mathbb{Z}$-module of finite rank and let $q : L \to \mathbb{Z}$ be an integral quadratic form. The *Clifford algebra* of the quadratic module (L, q) is defined by a usual universal property and can be realized explicitly as

$$C(L, q) = T(L)/J(q), \quad T(L) = \bigoplus_{r \geq 0} L^{\otimes r}, \quad L^{\otimes 0} = \mathbb{Z},$$

where $T(L)$ is the tensor algebra of L, and $J(q)$ is the two-sided ideal of $T(L)$ generated by all elements of the form $x \otimes x - q(x) \cdot 1_T$, $x \in L$. For details, we

refer to [Knu91, Chap. IV] and [Bou59, Chap. IX, §9]. The Clifford algebra C is an associative algebra with 1, equipped with a homomorphism $i : L \to C$ of $\mathbb{Z}$-modules such that

$$[i(x)]^2 = q(x) \cdot 1_C \quad \text{for all } x \in L,$$

and such that for any associative ring B, any $\mathbb{Z}$-linear map $L \to B$ satisfying the same quadratic relation factors through i.

Since $L \hookrightarrow T(L)$, we may occasionally denote the image of $x \in L$ in $C(L, q)$ by x. Observe that $C(L, q)$ is generated, as a $\mathbb{Z}$-algebra, by the image of L. Indeed, let $j : L \to C$ be the above map. The degree-zero inclusion gives multiplication in the tensor algebra $T(L)$ as $v_1 \otimes \cdots \otimes v_r = v_1 \cdots v_r$. Every element of $T(L)$ is a linear combination of such products (including the empty product), so the same holds in the quotient, i.e., in the Clifford algebra.

The universal property of the Clifford algebra gives us the following.

Lemma 7.2. *Let R be a ring, and let $f : L \to R$ be a $\mathbb{Z}$-linear map such that $f(x)^2 = q(x) \cdot 1_R$ for all $x \in L$. Then there is a unique ring homomorphism $\tilde{f} : C(L, q) \to R$ with $\tilde{f}(x) = f(x)$ for $x \in L$, and its image is the associative algebra generated by $f(L)$ over $\mathbb{Z}$. We denote this $\mathbb{Z}$-algebra by $\mathbb{Z}\langle f(L)\rangle$.*

Proof. This follows from the universal property of $T(L)$. The homomorphism f extends to a ring homomorphism $T(L) \to R$ whose kernel contains the generators of $J(q)$, hence it factors through the Clifford algebra $C(L, q)$. Since $C(L, q)$ is generated as a ring by the image of L, the extension is unique and its image is the subring generated by $f(L)$.

7.2 The Clifford algebra of the rank-five Rosati lattice $\mathscr{S}_A$

We apply this to $L = \mathscr{S}_A$, $R = \mathrm{End}(A)$, and the inclusion map $\mathscr{S}_A \hookrightarrow \mathrm{End}(A)$, which satisfies $x^2 = q_A(x) \cdot 1_{\mathrm{End}(A)}$ by (3). We obtain a unique ring homomorphism

$$\varphi_A : C(\mathscr{S}_A, q_A) \to \mathrm{End}(A), \quad \varphi_A(x) = x \quad (x \in \mathscr{S}_A), \tag{5}$$

which we call the *Clifford homomorphism*. Thus, we have

$$\varphi_A\big(C(\mathscr{S}_A, q_A)\big) = \mathbb{Z}\langle \mathscr{S}_A \rangle. \tag{6}$$

Since κ_A maps $\mathscr{S}_A$ to S_h, it maps the image of φ_A onto $\mathbb{Z}\langle S_h \rangle$. Thus, our problem is reduced to a pure matrix algebra problem. We need to show that

$$\mathbb{Z}\langle S_h \rangle = \mathrm{M}_2(\mathcal{O}).$$

We will prove this by proving that this holds at every prime.

For a prime ℓ, let

$$S_{h,\ell} = \{X \in \mathrm{M}_2(\mathcal{O}_\ell) \mid X^* h = hX,\ \mathrm{Trd}(X) = 0\}.$$

We use the following basic local-global argument.

Lemma 7.3. *Let M and N be $\mathbb{Z}$-lattices and let $M \subset N$. If $M \otimes_{\mathbb{Z}} \mathbb{Z}_\ell = N \otimes_{\mathbb{Z}} \mathbb{Z}_\ell$ for every prime ℓ, then $M = N$.*

This is classical, see, e.g., [Voi21, 9.4.6 and 9.5.3]. We can work directly with $S_{h,\ell}$, since it is easily seen to be the extension of scalars of S_h to $\mathbb{Z}_\ell$.

Lemma 7.4. *For every prime ℓ, we have $S_h \otimes \mathbb{Z}_\ell = S_{h,\ell}$, hence*

$$\mathbb{Z}\langle S_h \rangle \otimes_{\mathbb{Z}} \mathbb{Z}_\ell = \mathbb{Z}_\ell \langle S_{h,\ell} \rangle,$$

as submodules of $\mathrm{M}_2(\mathcal{O}) \otimes \mathbb{Z}_\ell = \mathrm{M}_2(\mathcal{O}_\ell)$. Thus, $\mathbb{Z}\langle S_h \rangle = \mathrm{M}_2(\mathcal{O})$ if and only if $\mathbb{Z}_\ell \langle S_{h,\ell} \rangle = \mathrm{M}_2(\mathcal{O}_\ell)$ for every prime ℓ.

Proof. Since $\mathbb{Z}\langle S_h \rangle \otimes \mathbb{Z}_\ell$ is the $\mathbb{Z}_\ell$-span of the products $s_1 \cdots s_m$ with $s_i \in S_h$, and every element of $S_{h,\ell} = S_h \otimes \mathbb{Z}_\ell$ is a $\mathbb{Z}_\ell$-linear combination of elements of S_h, we see that the two modules coincide.

We need to verify this for every prime ℓ. When ℓ is unramified, that is, $\ell \neq p$, we have $\mathcal{O}_\ell = \mathrm{M}_2(\mathbb{Z}_\ell)$, and when $\ell = p$, $\mathcal{O}_\ell$ is the unique maximal order in the division quaternion algebra D over $\mathbb{Z}_\ell$.

Proposition 7.5. *For every prime ℓ, there exists $W \in \mathrm{GL}_2(\mathcal{O}_\ell)$ such that*

$$W^* h W = H, \quad \text{where } H = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Observe that this formulation combines both the ramified and the unramified case. As such, the off-diagonal element 1 here denotes the identity of the ring $\mathcal{O}_\ell$, which via an appropriate isomorphism $\mathcal{O}_\ell \xrightarrow{\sim} \mathrm{M}_2(\mathbb{Z}_\ell)$ in the unramified case, is the identity matrix I_2.

Proof. For $\ell = p$, this is exactly [Pha26, Lemma 3.8(i)]. The unramified case also follows upon a re-ordering of the symplectic bases. Indeed, by [Pha26, Lemma 3.4], we have a $\mathbb{Z}_\ell$-algebra isomorphism

$$\rho : \mathrm{M}_2(\mathcal{O}_\ell) \xrightarrow{\sim} \mathrm{M}_4(\mathbb{Z}_\ell),$$

such that

$$\rho(X^*) = J_0^{-1} \rho(X)^T J_0, \quad J_0 = \begin{pmatrix} J_2 & 0 \\ 0 & J_2 \end{pmatrix}, \quad J_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

For a hermitian matrix $g \in \mathrm{GL}_2(\mathcal{O}_\ell)$, let

$$\Omega_g = J_0 \rho(g).$$

Observe that

$$\Omega_H = \begin{pmatrix} J_2 & 0 \\ 0 & J_2 \end{pmatrix} \cdot \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix} = \begin{pmatrix} 0 & J_2 \\ J_2 & 0 \end{pmatrix}.$$

Then, for any $W \in \mathrm{GL}_2(\mathcal{O}_\ell)$, we have

$$\begin{aligned}\rho(W^* g W) &= J_0^{-1}\rho(W)^T J_0 \rho(g)\rho(W) \\ &= J_0^{-1}\rho(W)^T \Omega_g \rho(W).\end{aligned}$$

Thus,

$$W^* h W = H \quad \Leftrightarrow \quad \rho(W)^T \Omega_h \rho(W) = \Omega_H.$$

The matrix Ω_h is nonsingular and alternating, and so is Ω_H. Thus, there is a matrix $U \in \mathrm{GL}_4(\mathbb{Z}_\ell)$ such that $U^T \Omega_h U = \Omega_H$. Thus, the matrix $W = \rho^{-1}(U)$ satisfies $W^* h W = H$.

Lemma 7.6. *Let ℓ be a prime and let $W \in \mathrm{GL}_2(\mathcal{O}_\ell)$ be such that $W^* h W = H$ as in the previous proposition. Then, the map $\gamma_W : X \mapsto W^{-1} X W$ is an automorphism of the ring $\mathrm{M}_2(\mathcal{O}_\ell)$ and maps $S_{h,\ell}$ onto*

$$\begin{aligned}S_{H,\ell} &= \{Y \in \mathrm{M}_2(\mathcal{O}_\ell) \mid Y^* H = HY,\ \mathrm{Trd}(Y) = 0\} \\ &= \left\{ \begin{pmatrix} z & b \\ c & -z \end{pmatrix} \mid z \in \mathcal{O}_\ell^0,\ b, c \in \mathbb{Z}_\ell \right\},\end{aligned}$$

where $\mathcal{O}_\ell^0$ denotes the set of elements of $\mathcal{O}_\ell$ of reduced trace zero.

Proof. When $\ell = p$, the result is a special case of [Pha26, Lemma 3.3 and (3.11)]. For $\ell \neq p$, this is contained in the proof of [Pha26, Proposition 3.6 and (3.10)].

It remains to prove the following ℓ-adic result.

Proposition 7.7. *For every prime ℓ, $\mathbb{Z}_\ell\langle S_{H,\ell}\rangle = \mathrm{M}_2(\mathcal{O}_\ell)$.*

Before proving this, we need the one-dimensional analogue of this result, namely, that $\mathbb{Z}_\ell\langle \mathcal{O}_\ell^0\rangle = \mathcal{O}_\ell$. This is in fact true globally, and the local case will follow.

Lemma 7.8. *There exist $x, y \in \mathcal{O}^0$ such that $\mathrm{trd}(xy)$ is odd, and for any such elements,*

$$\mathcal{O} = \mathbb{Z} \cdot 1 + \mathcal{O}^0 + \mathbb{Z} \cdot (xy).$$

In particular, $\mathbb{Z}\langle \mathcal{O}^0\rangle = \mathcal{O}$. As a consequence, $\mathbb{Z}_\ell\langle \mathcal{O}_\ell^0\rangle = \mathcal{O}_\ell$.

Proof. We have an isomorphism $\mathcal{O}/2\mathcal{O} \cong \mathrm{M}_2(\mathbb{F}_2)$. Indeed, since 2 is unramified in B, $B \otimes \mathbb{Q}_2 \cong \mathrm{M}_2(\mathbb{Q}_2)$ and $\mathcal{O} \otimes \mathbb{Z}_2$ is a maximal $\mathbb{Z}_2$-order in $B \otimes \mathbb{Q}_2$, hence conjugate to $\mathrm{M}_2(\mathbb{Z}_2)$, so $\mathcal{O} \otimes \mathbb{Z}_2 \cong \mathrm{M}_2(\mathbb{Z}_2)$. Reducing modulo 2 locally while extending scalars globally gives the (not all canonical) isomorphisms

$$\mathrm{M}_2(\mathbb{F}_2) \cong (\mathcal{O} \otimes \mathbb{Z}_2)/2(\mathcal{O} \otimes \mathbb{Z}_2) \cong \mathcal{O} \otimes (\mathbb{Z}_2/2\mathbb{Z}_2) \cong \mathcal{O} \otimes \mathbb{F}_2 \cong \mathcal{O}/2\mathcal{O}.$$

Under the isomorphism $\mathcal{O}/2\mathcal{O} \cong \mathrm{M}_2(\mathbb{F}_2)$, the reduced trace becomes the ordinary matrix trace. The matrix units E_{12} and E_{21} have trace zero, and $E_{12}E_{21} = E_{11}$ which has trace 1. Since trd is surjective on $\mathcal{O}$, there exists $a \in \mathcal{O}$ with

$\mathrm{trd}(a) = 1$. For any $z \in \mathcal{O}$ with even trace, $z - \mathrm{trd}(z)a \in \mathcal{O}^0$ is congruent to z modulo $2\mathcal{O}$. Thus, from E_{12} and E_{21} with zero trace mod 2, we can lift them to an element of even reduced trace, which we then correct to an element of zero reduced trace. If $x, y \in \mathcal{O}^0$ are such lifts of E_{12}, E_{21}, then $\mathrm{trd}(xy) \equiv 1 \pmod 2$. Now, let $m = \mathrm{trd}(xy)$ and $t = xy - \frac{1}{2}(m-1) \cdot 1 \in \mathbb{Z}(xy) + \mathbb{Z} \cdot 1$. Observe that $(m-1)/2$ is an integer since m is odd. Then, $\mathrm{trd}(t) = 1$ and $t \in \mathbb{Z} \cdot 1 + \mathbb{Z} \cdot (xy)$. Since $\mathrm{trd}(t) = 1$, we can proceed as before and conclude that for $z \in \mathcal{O}$, we can write

$$z = \mathrm{trd}(z)t + (z - \mathrm{trd}(z)t), \quad z - \mathrm{trd}(z)t \in \mathcal{O}^0.$$

Thus, $\mathcal{O} = \mathbb{Z} \cdot 1 + \mathcal{O}^0 + \mathbb{Z} \cdot (xy)$. Tensoring with $\mathbb{Z}_\ell$ gives the corresponding local result.

We can now complete the proof of Proposition 7.7 and, as a consequence, that of Theorem 7.1.

Proof (of Proposition 7.7). Let $R = \mathbb{Z}_\ell\langle S_{H,\ell}\rangle$. Let $e_{ij} \in \mathrm{M}_2(\mathcal{O}_\ell)$, $1 \le i, j \le 2$, denote the matrix units. From the explicit description

$$S_{H,\ell} = \left\{ \begin{pmatrix} z & b \\ c & -z \end{pmatrix} \mid z \in \mathcal{O}_\ell^0,\, b, c \in \mathbb{Z}_\ell \right\},$$

we see that $e_{12}, e_{21} \in S_{H,\ell}$. Hence, $e_{11} = e_{12}e_{21} \in R$ and similarly, $e_{22} = e_{21}e_{12} \in R$. Hence, $\mathrm{M}_2(\mathbb{Z}_\ell) \subset R$.

If $z \in \mathcal{O}_\ell^0$ has reduced trace zero, then

$$\begin{pmatrix} z & 0 \\ 0 & -z \end{pmatrix} \in S_{H,\ell}, \quad \begin{pmatrix} z & 0 \\ 0 & -z \end{pmatrix} (e_{11} - e_{22}) = zI_2,$$

so R contains $\mathbb{Z}_\ell\langle \mathcal{O}_\ell^0 \rangle I_2 = \mathcal{O}_\ell I_2$ by Lemma 7.8. For $a \in \mathcal{O}_\ell$, the matrix $(aI_2)e_{ij}$ is the matrix with a in position (i, j) and 0 elsewhere. Since these span $\mathrm{M}_2(\mathcal{O}_\ell)$, this completes the proof.

The proof of Theorem 7.1 is complete. □

Corollary 7.9. *Assume Heuristics 1 and 2. Let $A/\mathbb{F}_{p^2}$ be a maximal superspecial principally polarized abelian surface. Then one can compute a generating set for* $\mathrm{End}(A)$ *as a $\mathbb{Z}$-module, consisting of at most* 32 *efficiently represented polarized endomorphisms, in expected time and memory $p^{4/5+o(1)}$.*

Proof. This follows immediately by combining Theorem 6.1, Proposition 6.2, and Theorem 7.1, by noting that the (products of the) generators for $\mathscr{S}_A$ are automatically polarized.

8 The EndRing and Isogeny problems

From the generating set for $\mathrm{End}(A)$ from Corollary 7.9, we can immediately deduce, thanks to the algorithms of Page–Robert–Soumier [PRS26], several additional consequences. The first is a solution to EndRing.

Theorem 8.1. *Assume Heuristics 1 and 2. Let $A/\mathbb{F}_{p^2}$ be a maximal superspecial principally polarized abelian surface. Then we can compute a $\mathbb{Z}$-basis for* $\mathrm{End}(A)$ *consisting of efficient representations, together with a multiplication table for the ring structure, and a matrix representation for the Rosati involution, in expected time and memory $p^{4/5+o(1)}$.*

Proof. Use Corollary 7.9 to obtain a generating set for $\mathrm{End}(A)$ as a $\mathbb{Z}$-module. By [PRS26, Thm. 5.8], we can turn this into a $\mathbb{Z}$-basis together with a multiplication table in expected time $(\log p)^{O(1)}$. Recovering the matrix of the Rosati involution then follows from [PRS26, Example 5.9].

8.1 From the endomorphism ring to effective IKO coordinates

Following [PRS26] further, fix the reference supersingular elliptic curve E_0 with the known action of $\mathcal{O}_0 = \mathrm{End}(E_0)$ [PRS26, §2.1]. We write λ_0 for the product principal polarization on E_0^2. For a matrix $T = (t_{ij}) \in \mathrm{M}_2(\mathcal{O}_0)$, we let $[T]_0$ denote the induced endomorphism of E_0^2, namely, it is given by evaluation on points (P, Q) by

$$[T]_0(P,Q) = \big(t_{11}(P) + t_{12}(Q), t_{21}(P) + t_{22}(Q)\big).$$

We use the explicit operations described in [PRS26, §5].

Theorem 8.2 (GRH). *Let (A, λ_A) be a maximal superspecial principally polarized abelian surface over $\mathbb{F}_{p^2}$. Given a $\mathbb{Z}$-basis of* $\mathrm{End}(A)$ *as in Theorem 8.1, we can compute a positive definite Hermitian matrix of Hermitian determinant one, $h_A \in \mathrm{M}_2(\mathcal{O})$, and mutually inverse unpolarized isomorphisms*

$$\iota_A : A \xrightarrow{\sim} E_0^2, \quad \iota_A^{-1} : E_0^2 \xrightarrow{\sim} A,$$

which can be evaluated explicitly, such that

$$(\iota_A^{-1})^\vee \circ \lambda_A \circ \iota_A^{-1} = \lambda_0 \circ [h_A]_0. \tag{7}$$

The expected running time is polynomial in $\log(p)$.

Proof. By [PRS26, Proposition 4.11], we obtain an isomorphism $\mathrm{End}(A) \xrightarrow{\sim} \mathrm{M}_2(\mathcal{O}_0)$. This algorithm, under GRH, has expected running time polynomial in $\log(p)$. By [PRS26, Theorem 4.12], under GRH, we also obtain abstract (resp. standard) module representation and refined abstract (resp. standard) endomorphism ring representation, together with conversions between these, with the same complexity.

Although there may be two actual candidate modules because of A and its Frobenius conjugate, by [PRS26, Remark 5.10], they can be distinguished. Thus, we have an effective hermitian module $M_A = \mathcal{O}_0^2$ with hermitian form $H_A(u, v) = uS_Av^*$. By [PRS26, Theorem 6.8 and Corollary 6.10], we can construct an unpolarized isomorphism $\iota_A : A \xrightarrow{\sim} E_0^2$. For $\varphi_1, \varphi_2 \in \mathrm{Hom}(A, E_0)$, the hermitian form is $H(\varphi_1, \varphi_2) = \varphi_1 \circ \lambda_A^{-1} \circ \varphi_2^\vee \circ \lambda_{E_0} \in \mathcal{O}_0$. Write $\varphi_1 \circ \iota_A^{-1} = (u_1, u_2) = u$

and $\varphi_2 \circ \iota_A^{-1} = (v_1, v_2) = v$ as row vectors. Then, u sends a point $\left(\begin{smallmatrix} P_1 \\ P_2 \end{smallmatrix}\right) \in E_0^2$ to the point $u_1(P_1) + u_2(P_2) \in E_0$, similarly for v. The hermitian form becomes $uh_A^{-1}v^*$, hence we get

$$H_A(u, v) = uS_Av^* \in \mathcal{O}_0, \quad S_A = h_A^{-1}.$$

Therefore, the sought after IKO matrix is $h_A = S_A^{-1}$. This completes the proof.

8.2 Computing and evaluating a connecting matrix

Corollary 8.3 (GRH). *Let (A, λ_A) and (C, λ_C) be maximal principally polarized abelian surfaces. Assume that we are given the endomorphism rings* $\mathrm{End}(A)$ *and* $\mathrm{End}(C)$ *as in Theorem 8.1, we can compute an efficient representation of a separable polarized isogeny $f : A \to C$ such that*

$$f^\dagger f = [N]_A, \quad 1 \leq N \leq \lfloor p/\sqrt{2} \rfloor, \quad \deg f = N^2,$$

in expected polynomial time in $\log(p)$ *and in the size of the representations.*

Proof. We apply Theorem 8.2 using the same E_0 and $\mathcal{O}_0$ for both surfaces. We obtain IKO matrices h_A and h_C. We then apply the deterministic algorithm of [Pha26] to h_A and h_C to obtain a connecting matrix. The algorithm returns a matrix $\gamma \in \mathrm{M}_2(\mathcal{O}_0)$ and a positive integer N such that $1 \leq N \leq \lfloor p/\sqrt{2} \rfloor$, with $\gamma^* h_C \gamma = N h_A$. We define

$$f = \iota_C^{-1} \circ [\gamma]_0 \circ \iota_A.$$

For a point P of A, we evaluate $\iota_A(P) = (P_1, P_2) \in E_0^2$. We apply the matrix γ to the column vector $\iota_A(P)$ using the known $\mathcal{O}_0$-action, and then apply and evaluate ι_C^{-1} on the resulting point. This defines a homomorphism of abelian varieties $A \to C$, functorially, on points. Then, equation (7) implies

$$\begin{aligned} f^\vee \lambda_C f &= (\iota_A^\vee [\gamma]_0^\vee (\iota_C^{-1})^\vee) \lambda_C (\iota_C^{-1} [\gamma]_0 \iota_A) \\ &= \iota_A^\vee [\gamma]_0^\vee (\lambda_0 [h_C]_0) [\gamma]_0 \iota_A \\ &= \iota_A^\vee \lambda_0 [\gamma^*]_0 \, [h_C]_0 \, [\gamma]_0 \iota_A \\ &= \iota_A^\vee \lambda_0 [\gamma^* h_C \gamma]_0 \iota_A \\ &= \iota_A^\vee \lambda_0 [N h_A]_0 \iota_A \\ &= N \lambda_A. \end{aligned}$$

Therefore, $f^\dagger f = [N]_A$ as desired. The remaining conditions on f are routinely verified (see [Pha26] for instance), and computing the connecting matrix algorithm of [Pha26] has polynomial cost. The composition has expected polynomial cost [PRS26], and this completes the proof.

We thus conclude with an algorithm for Isogeny.

Theorem 8.4. *Assume Heuristics 1 and 2 and GRH. Let A, C be maximal superspecial principally polarized abelian surfaces over $\mathbb{F}_{p^2}$. Then we can compute an efficient representation of a polarized isogeny $A \to C$ in expected time and memory $p^{4/5+o(1)}$.*

Proof. Compute both endomorphism rings using Theorem 8.1, then, assuming GRH, obtain the IKO matrices via Theorem 8.2. Then return the isogeny from Corollary 8.3. Alternatively, one can use KLPT^2 [CDK$^+$25] to obtain an isogeny with smooth multiplier.

9 Experiments

We now report on experimental results, mostly geared towards Heuristic 1. All experiments were performed using SageMath [The26b]. All data as well as the code used for generating this data can be found in the git repository associated to this paper.[4]

Setup

Let p be a prime number. For our experiments, we used the smallest prime $p \equiv 7 \pmod 8$ such that $p > 2^b$, where $b \in \{128, 256\}$.

Using a modified version of the function RandomPolarization from the git repository[5] associated to the KLPT^2 paper [CDK$^+$25], we generate random IKO matrices g_A corresponding to superspecial principally polarized abelian surfaces $A/\mathbb{F}_{p^2}$. Next, we compute the associated dimension-5 quadratic form q_A from Equation (2), together with the smallest integer n represented by it. For the latter, we used the PARI/GP-function [The26a] qfminim, with flag = 2. We then compare the observed B-smoothness probability distribution of the integer n with what one expects for random integers of the same size.

More precisely, we compare with the B-smoothness probability distribution of a *random* integer $n \in S_p := \{1, \dots, \lfloor (p^2/2)^{1/5} \rfloor\}$, where *random* is according to three possible probability distributions on S_p. These three distributions are obtained by assigning to $n \in S_p$ a (suitably normalized) weight of either:

(i) 1 (corresponding to the uniform distibution);
(ii) $a_2(n)$ (the distribution one would expect if α as in Theorem 3.3 were a random (n, n)-isogeny with domain A);
(iii) $b_2(n)$ (the distribution one would expect if α as in Theorem 3.3 were a random polarized isogeny with domain A),

where $a_2(n)$ and $b_2(n)$ are as in Section 4. Set $X := \lfloor (p^2/2)^{1/5} \rfloor$. The B-smoothness probability distributions (as a function of B) corresponding to each of the three cases above are, respectively: $\delta(X, B) = \Psi(X, B)/X$; $\sigma(X, B) := \mathcal{A}_2(X, B)/\mathcal{A}_2(X, X)$; and $\tau(X, B) := \mathcal{B}_2(X, B)/\mathcal{B}_2(X, X)$, where

$$\mathcal{A}_2(X, B) := \sum_{\substack{1 \le n \le X \\ P^+(n) \le B}} a_2(n), \qquad \text{and} \qquad \mathcal{B}_2(X, B) := \sum_{\substack{1 \le n \le X \\ P^+(n) \le B}} b_2(n).$$

[4] https://github.com/houbenmr/attack-four-fifth
[5] https://github.com/KLPT2/KLPT2

Remark 9.1. Our sampling method for g_A does not necessarily yield an IKO matrix corresponding to a uniformly random principally polarized abelian surface $A/\mathbb{F}_{p^2}$. In fact, during our implementations of these experiments, we noticed that the original function RandomPolarization from KLPT^2 generates IKO matrices corresponding to abelian surfaces with exceptionally small endomorphisms, with multipliers on the order of $\sim p^{1/4}$. This can be explained by the observation that, through its use of the RepresentInteger function[6], the generated surfaces lie, with high probability, at a small distance from an $\mathbb{F}_p$-rational surface. We removed this bias and adjusted the code accordingly. The uniform randomness of the IKO matrix up to equivalence remains heuristic.

Results of the experiments

Smoothness probabilities. For the 128- and 256-bit primes, we sampled 10000 random IKO matrices each. We factored the associated minimal multiplier n and recorded its largest prime factor B. The functions $\delta(X, B), \sigma(X, B)$, and $\tau(X, B)$ were approximated using 10^6 samples each. The resulting B-smoothness probability distributions can be found in Figures 1–4.

The observed smoothness distributions support Heuristic 1. This does not yield a quantified statistical claim that the heuristic is correct. At small values of B, the distribution $\sigma(X, B)$ appears to be the closest fit. This a posteriori makes sense, considering Theorem 3.3, although a true bias or convergence towards this distribution cannot be concluded from the current data.

The size of the minimum multiplier. For the 128-bit prime, where $\log_2(X) \approx 51$, the $\log_2$ of the maximum, minimum, and median observed values of n were $50.66, 45.46$, and 49.60 respectively. For the 256-bit prime, $\log_2(X) \approx 102$, and the $\log_2$ of the maximum, minimum, and median of observed n were $101.85, 95.14$, and 100.80 respectively. This suggests that the upper bound of $n \leq X = \lfloor (p/2) \rfloor^{1/5}$ is close to tight (although it does not exclude a small possible improvement). The minimum observed value of n is larger than one might expect from either of the three distributions on S_p mentioned above (one experiment sampling 10000 values for each yielded minima of $88.30, 87.43$, and 83.63, respectively), possibly suggesting a bias for n to be close to the general upper bound of X. Further experimental evidence should be obtained before these observations can be rigorously established.

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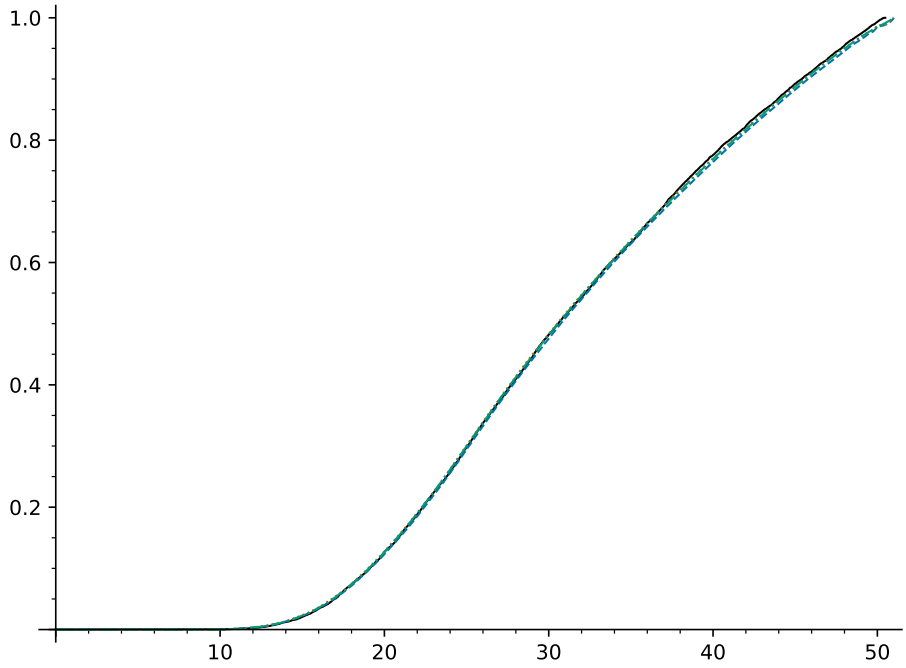


Fig. 1. For a 128-bit prime p: the B-smoothness probability distribution (black) of the smallest integer n represented by q_A, together with $\delta(X, B)$ (green, dash-dot), $\sigma(X, B)$ (blue, dashed), and $\tau(X, B)$ (orange, dotted). The x-axis displays $\log_2(B)$.

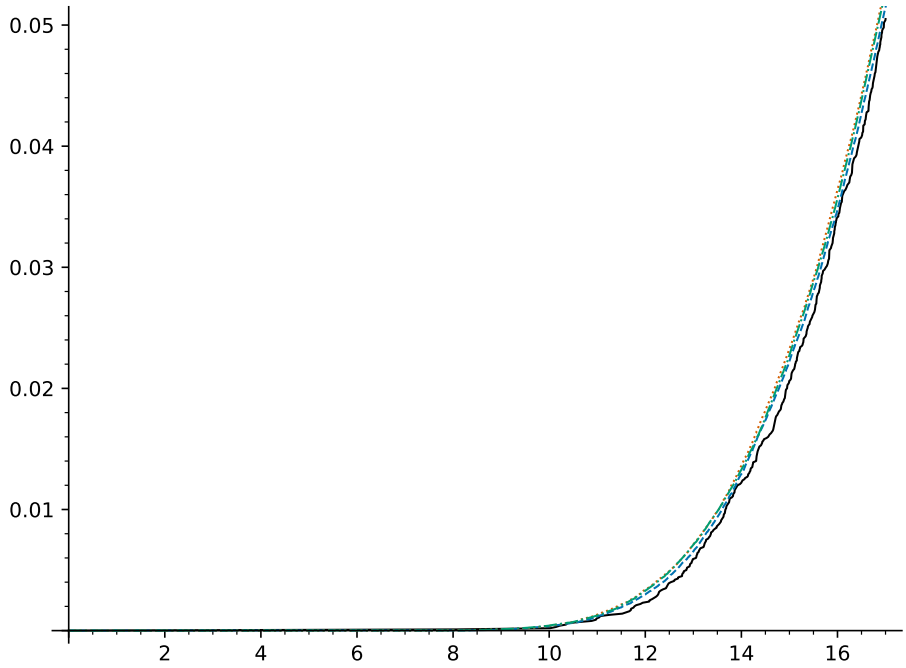


Fig. 2. A zoom-in on the same plot for the 500 smoothest samples of n.

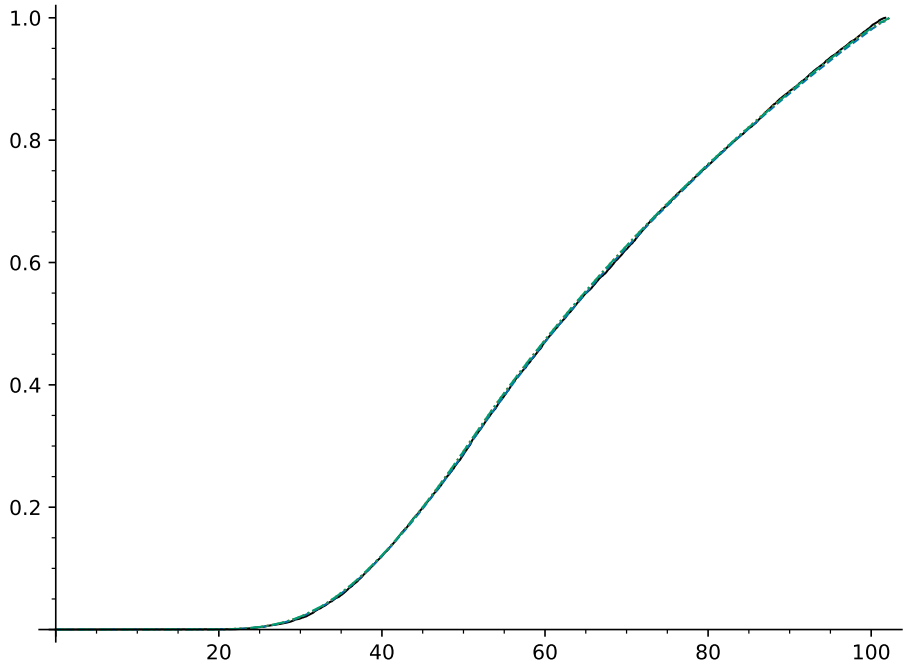


Fig. 3. The same as Figure 1 for a 256-bit prime.

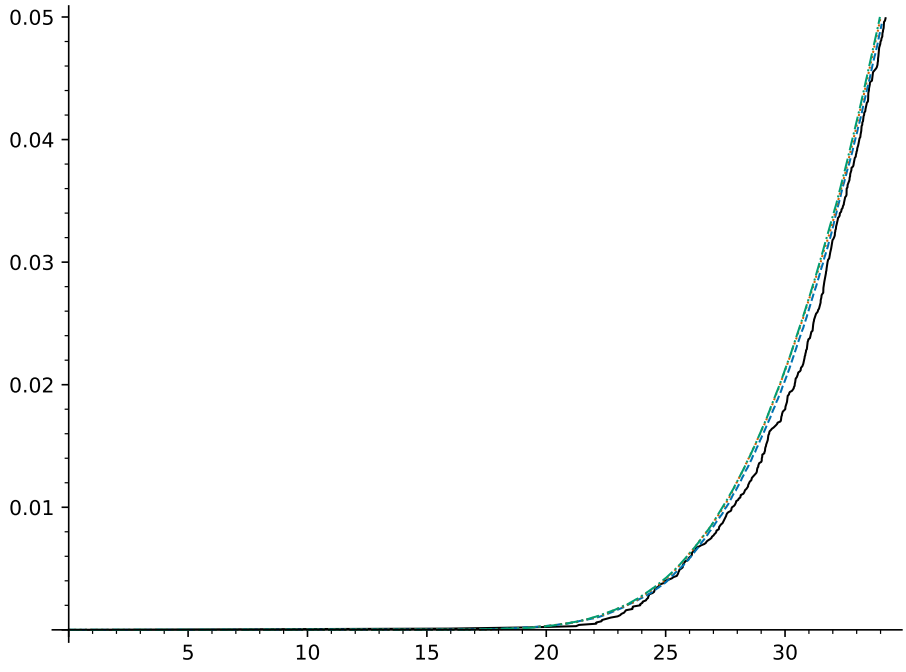


Fig. 4. A zoom-in on the 500 smoothest samples of n.